NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 21/01/2026
Time: 3 Hrs | GTM – 21 | Max. Marks: 300

KEY SHEET

PHYSICS

01	02	03	04	05	06	07	08	09	10
D	C	A	B	B	B	D	A	C	C
11	12	13	14	15	16	17	18	19	20
C	A	C	C	D	C	A	D	C	A
21	22	23	24	25					
100	6	30	1	16					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
C	B	D	C	D	A	C	A	B	A
36	37	38	39	40	41	42	43	44	45
C	D	D	B	A	B	B	B	C	D
46	47	48	49	50					
5	6	29	9	40					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	B	B	A	C	D	B	A	B
61	62	63	64	65	66	67	68	69	70
A	C	B	A	A	B	B	B	C	A
71	72	73	74	75					
16	6	3	9	2					

Rankers Academy JEE

or More Join: @IITJEE_Advanc

HINTS
PHYSICS

1. A : Since polar dielectric are randomly oriental $\vec{P}_{net} = \vec{0}$.

   R : If $\vec{E}$ is absent, polar dielectric remain polar and are randomly oriented.

2. Current sensitivity $= \frac{\theta}{I} = \frac{NAB}{c}$

4. $T \times 4\pi\left(R^2 - Nr^2\right) = \left(\frac{4}{3} \times R^3 \times \rho\right) s\Delta\theta$

$$\Delta\theta = \frac{T \times 4\pi\left(R^2 - Nr^2\right)}{\frac{4}{3}\pi R^3 \rho \times s}$$

$$= \frac{3T}{\rho s}\left[\frac{R^2}{R^3} - \frac{Nr^2}{R^3}\right]$$

$$= \frac{3T}{\rho s}\left[\frac{1}{R} - \frac{\left(R^3/r^3\right) \times r^2}{R^3}\right]$$

$$= \frac{3T}{\rho s}\left[\frac{1}{R} - \frac{1}{r}\right]$$

5. Initial de-Broglie wavelength of electron $\lambda_0 = \frac{h}{mv_0}$

Force on electron in electric field $F = -eE = -e\left[-E_0 i\right] = eE_0 i$

Acceleration of electron $a = \frac{F}{m} = \frac{eE_0 i}{m}$

Velocity of electron after time t.

$$v = v_0 i + \left(\frac{eE_0 i}{m}\right) t = \left(v_0 + \frac{eE_0 t}{m}\right)\hat{i}$$

$$= v_0\left(1 + \frac{eE_0 t}{mv_0}\right)\hat{i}$$

De-Broglie wavelength associated with electron at time t is

$$\lambda = \frac{h}{mv}$$

$$\Rightarrow = \frac{h}{m\left[v_0\left(1 + \frac{eE_0 t}{mv_0}\right)\right]} = \frac{\lambda_0}{\left[1 + \frac{eE_0 t}{mv_0}\right]} \qquad \left[\because \lambda_0 = \frac{h}{mv_0}\right]$$

6. Amplitude of reflected wave $A_r = \frac{2}{3} \times A_i = \frac{2}{3} \times 0.6 = 0.4$ units

For More Join: @IITJEE_Advanc

Given equation of incident wave $y_r = 0.6 \sin 2\pi\left(t - \frac{x}{2}\right)$

Equation of reflected wave is $y_r = A_r \sin 2\pi\left(t + \frac{x}{2} + \pi\right)$

[$\because$ At denser medium, phase changes by π]

The positive sign is due to reversal of direction of propagation

So, $y_r = -0.4 \sin 2\pi\left(t + \frac{x}{2}\right)$ $\left[\because \sin(\pi + \theta) = -\sin\theta\right]$

7. Given, fundamental quantities are momentum (p), area (A) and time (T).

We can write energy E as

$$E \propto p^a A^b T^c$$

$$E = kp^a A^A T^c$$

Where k is dimensionless constant of proportionality.

Dimensions of $E = [E] = [ML^2T^{-2}]$ *and* $[p] = [MLT^{-1}]$

$$[A] = [L^2]$$

[T] = [T]

$$[E] = [K][p]^a [A]^b [T]^c$$

Putting all the dimensions, we get $ML^2T^{-2} = [MLT^{-1}]^a [L^2]^b [T]^c$

$$= M^a L^{2b+a} T^{-a+c}$$

By principal of homogeneity of dimensions

$$a = 1,\ 2b + a = 2$$

$\Rightarrow \quad 2b + 1 = 2$

$\Rightarrow \quad b = 1/2 - a + c = -2$

$\Rightarrow \quad c = -2 + a = -2 + 1 = -1$

Hence, $E = pA^{1/2} T^{-1}$

8. $\frac{U}{A} = \frac{B^2}{2\mu_r u_0} \Rightarrow U = \frac{B^2}{\mu_0} V$

9. The work done by a electrostatic force is given by $W_{12} = q(V_2 - V_1)$. Here initial and final potentials are same in all three cases and same charge is moved, so work done is same in all three cases.

10. a) When the rod is hammered, the external work is done on the rod which increases its temperature. The process cannot be retraced itself.

b) In this process energy in the form of heat is transferred to the gas in the small container by big reservoir at temperature T_2.

d) As the weight is added to the cylinder arrangement in the form of external pressure hence, it cannot be reversed back itself.

For More Join: @IITJEE_Advanc

11. The atomic number of lithium is 3, therefore the radius of Li^{++} ion in its ground state, on the basis of Bohr's model, will be about $\frac{1}{3}$ times to that of Bohr radius.

Therefore, the radius of lithium ion is near $\frac{53}{3} = 18\ pm$

13. $C = A.B \ \ and \ \ D = \bar{A}.B$

$E = C + D = (A.B) + (\bar{A}.B)$

14. $$\frac{1}{2\left(L - \frac{a}{2}\right)} - \frac{1}{\left(L - \frac{a}{2}\right)} = -\frac{1}{f} \quad \text{..... (1)}$$

$$-\frac{1}{2\left(L + \frac{a}{2}\right)} - \frac{1}{\left(L + \frac{a}{2}\right)} = -\frac{1}{f} \quad \text{..... (2)}$$

Solving we get a = L.

15. For a particle performing uniform circular motion

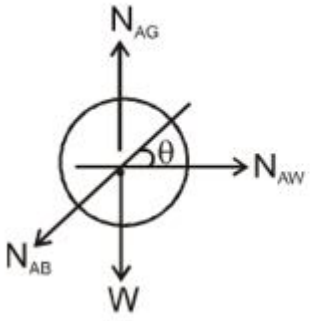
(i) speed will be constant throughout.

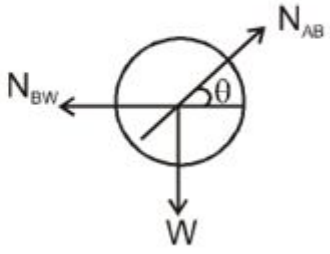
(ii) velocity will be tangential in the direction of motion at a particular point.

(iii) acceleration $a = \frac{v^2}{r}$ will always be towards centre of the circular path.

(iv) angular momentum (mvr) is constant in magnitude and direction/out of the plane perpendicular as well.

16. The FBD's of A and B are:





for the horizontal equilibrium of the system (A & B):

$N_{AW} = N_{BW}$ ..... (1)

And by diagrams:

$$\cos\theta = \frac{6}{10} \Rightarrow \theta = 53^0$$

Applying NLM on sphere B.

$\Rightarrow N_{AB}\sin\theta = w$

For More Join: @IITJEE_Advanc

And $N_{BW} = N_{AB}\cos\theta = \dfrac{w}{(4/5)} \times \dfrac{3}{5} = \dfrac{3w}{4}$

So $N_{AW} = N_{BW} = \dfrac{3w}{4}$

17. (A) Focal length is minimum in case-I, therefore power is maximum

18. $L.C. = \dfrac{0.5}{50} = 0.01mm$

$\text{Zero error} = (50-45) = 5\times 0.01 = 0.05mm(\text{Negative})$

$\text{Reading} = (0.5 + 25\times 0.01) + 0.05 = 0.80\,mm$

20. $I = I_{max}COS^2\left(\dfrac{\phi}{2}\right)$

$\Rightarrow \dfrac{3}{4}I_{max} = I_{max}COS^2\left(\dfrac{\phi}{2}\right)$

$\Rightarrow \dfrac{\phi}{2} = \dfrac{\pi}{2}$

$\Rightarrow \dfrac{\phi}{2} = \dfrac{\pi}{3}\left(\dfrac{2\pi}{\lambda}\right)(\mu-1)t$

$\Rightarrow t = \dfrac{\lambda}{6(\mu-1)} = \dfrac{6000}{6(1.5-1)} = 2000A^0$

21. $\tau = FR = 1\alpha \Rightarrow 40\times 0.1 = 0.4\,\alpha$

$\alpha = 10\,rad/s^2$

$W_f = 10\times 10 = 100\,rad/s$

22. $U = nC_vT = n\dfrac{5RT}{2}$

23. As $\mu = \dfrac{\sin\left(\dfrac{D_{\min}+A}{2}\right)}{\sin\left(\dfrac{A}{2}\right)}$

$\sqrt{2} = \dfrac{\sin\left(\dfrac{D_{\min}+60}{2}\right)}{\sin\left(\dfrac{60}{2}\right)}$

$\Rightarrow \dfrac{1}{\sqrt{2}} = \sin\left(\dfrac{D_{\min}+60}{2}\right) \Rightarrow \dfrac{D_{\min}+60}{2} = 45$

$\Rightarrow D_{\min} = 30$

For More Join: @IITJEE_Advanc

24. $T = K(\ell - \ell_0)$

$\Rightarrow 5 = K(1.4 - \ell_0)$

$\Rightarrow 7 = K(1.56 - \ell_0)$

$\Rightarrow \frac{5}{1.4 - \ell_0} = \frac{7}{1.56 - \ell_0}$

$\therefore \ell_0 = 1\,m$

25. $-\frac{GM_E m}{10R} + \frac{1}{2}mV_0^2 = -\frac{GM_E m}{R} + \frac{1}{2}mV^2$

$\frac{GM_E}{R}\left(1 - \frac{1}{10}\right) + \frac{V_0^2}{2} = \frac{V^2}{2}$

$V^2 = V_0^2 + \frac{9}{5}gR$

$V = \sqrt{V_0^2 + \frac{9}{5}gR} \approx 16\,km/s$

CHEMISTRY

26. Based on probability it is having n! possibilities.

27. B.P. () = 300.9

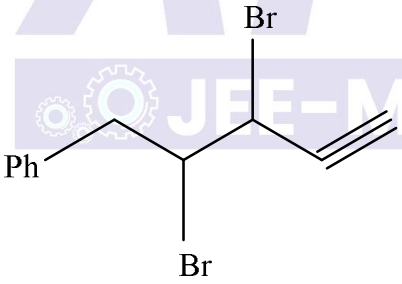
B.P. () = 309.1

28. $R - CN \rightarrow R - CHO$

29. H_2S - Acid

SO_2 - bleaching agent due to its reducing nature

30. $wt\%\ of\ N = \frac{28 \times 41.9}{22400 \times 0.3} \times 100$

31. 

32. $2P_x$ orbital having yz-nodel plane

For More Join: @IITJEE_Advanc

33. $\Delta G^0 = \left(\Delta E^0 - \Delta nRT\right) - T\Delta S^0$

34. Conceptual

35. Sp^3d^2 and Sp^3

36. d^2Sp^3 octahedral

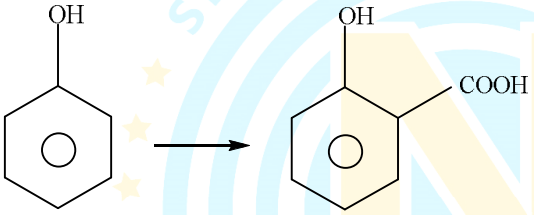
37. Distillation under reduced pressure is used for Glycerol.

38. XeF_4 square planar

39. Cyanide ion

40. Overal reaction endothermic

41.



42. On increase in pressure, equilibrium drive in backward.

$2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$

43.

$3.3\times10^{-3} - x$	$\frac{x}{2}$	$\frac{x}{2}$

44. $\left[H^+\right] = C\alpha \; ; \; K_a = C\alpha^2$

47. $\Delta Tf = Kf \times \frac{w}{mw} \times \frac{1000}{wt.\text{of solvent}}$

50. ksp of $Ag_2CrO_4 = 4x^3$

ksp of $AgBr = x^2$

For More Join: @IITJEE_Advanc

MATHEMATICS

51. $A+BA^T=I$ ..... (1)

$B+AB^T=I$ ..... (2)

$A^T+AB^T=I$ ..... (3)

From (2) and (3) $B=A^T$

$B^T=A \Rightarrow B=A^T$

From (2) $A^T+A^2=I$

$A^T=I-A^2$ ..... (4)

Now from (1) $A+\left(A^T\right)^2=I$

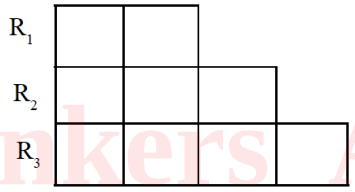
$A+\left(I-A^2\right)^2=I$

$A+I-2A^2+A^4=I$

$\therefore\ A^4-2A^2=-A$

$\therefore\ k=-1$

52.



R_1	R_2	R_3	
2	2	2	$\rightarrow 1\times3\times6=18$
2	1	3	$\rightarrow 1\times3\times4=12$
1	2	3	$\rightarrow 2\times3\times4=24$
2	3	1	$\rightarrow 1\times1\times4=4$
1	1	4	$\rightarrow 2\times3\times1=6$
1	3	2	$\rightarrow 2\times1\times6=12$
			76

53. $a+ar^{n-1}=66$ ..... (1)

$ar\,.\,ar^{n-2}=128$

$a^2\,.\,r^{n-1}=128$

$r^{n-1}=\frac{128}{a^2}$

From (1) $a+a\,.\,\frac{128}{a^2}=66$

For More Join: @IITJEE_Advanc

$a^2 - 66a + 128 = 0$

$(a-2)(a-64) = 0$

$a = 2,\ a = 64$

$\therefore\ 4 . r^{n-1} = 128$

$r^{n-1} = 32 = 2^5$

$n - 1 = 5 \Rightarrow n = 6$

$r = 2$

54. $g(x) = f^{-1}(x)$

$f(g(x)) = x$

$f^1(g(x))g^1(x) = 1$

$g^1(x) = \dfrac{1}{f^1(g(x))}$

$g^1(0) = \dfrac{1}{f^1(g(0))}$

$g^1(0) = \dfrac{1}{f^1(3)} = 11$

$f(3) = 0$

$f^{-1}(0) = 3$

$g(0) = 3$

$f^1(x) = \dfrac{1}{\sqrt{x^4 + 3x^2 + 13}}$

$f^1(3) = \dfrac{1}{\sqrt{81 + 27 + 13}}$

$= \dfrac{1}{\sqrt{121}} = \dfrac{1}{11}$

55. $I = 3\int_{-3}^{1} (x-k)(x+3)(x-1)\,dx = 64$ ..... (1)

$= 3\int_{-3}^{1} (-2-x-k)(-2-x+3)(-2-x-1)\,dx = 64$

$I = -3\int_{-3}^{1} (2+k+x)(x-1)(x+3)\,dx$ ..... (2)

$2I = 3\int_{-3}^{1} (x+3)(x-1)\{(x-k-2-k-x)\}\,dx$

For More Join: @IITJEE_Advanc

$$2I = 3(-2-2k)\int_{-3}^{1}\left(x^2+2x-3\right)dx$$

$$I = 3(-1-k)\left(\frac{x^3}{3}+x^2-3x\right)_{-3}^{1} = 64$$

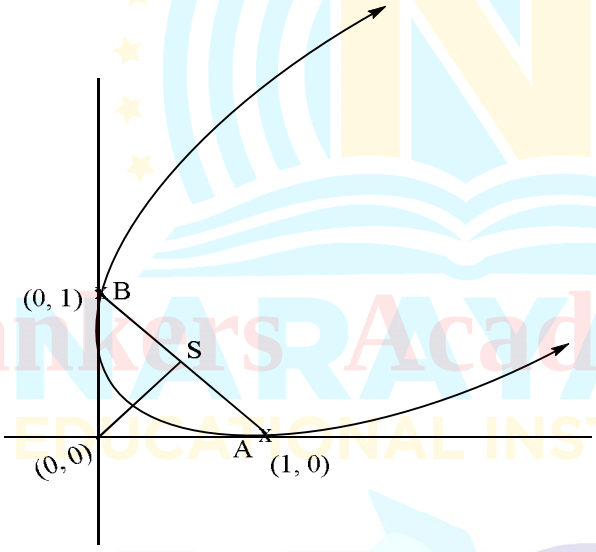
$$3(-k-1)\left[\frac{1}{3}+1-3+\frac{27}{3}-9-9\right] = 64$$

$$3(-k-1)\left[\frac{-32}{3}\right] = 64$$

$$32(k+1) = 64$$

$$k = 1$$

56.



Focus S is foot of perpendicular from (0, 0) onto $x+y=1$

$$\frac{h-0}{1} = \frac{k-0}{1} = \frac{-(-1)}{2}$$

$$h = \frac{1}{2},\ k = \frac{1}{2}$$

$$\therefore \text{ Focus } = S\left(\frac{1}{2},\frac{1}{2}\right)$$

$\therefore$ S is mid point of $AB \Rightarrow m=1,\ n=1$

$$m^2+n^2 = 2$$

57. $AC = \sqrt{15^2+20^2} = 25$

BD = 12

AD = 9

For More Join: @IITJEE_Advanc

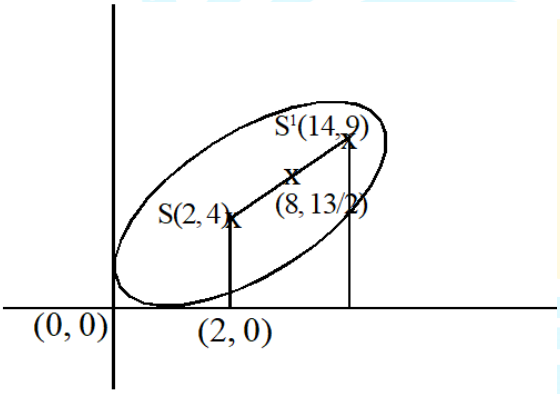
CD = 16

$$r_1 = \frac{\Delta}{S} = \frac{\frac{1}{2}\times 12\times 9}{\frac{12+9+15}{2}} \Rightarrow \frac{9\times 12}{36} = 3$$

$$r_1 = \frac{\Delta}{S} = \frac{\frac{1}{2}\times 16\times 12}{\frac{20+16+12}{2}} \Rightarrow \frac{16\times 12}{48} = 4$$

Distance $C_1C_2 = \sqrt{50}$

58.



$2ae = \sqrt{144+25} = 13$

$ae = \frac{13}{2}$ …….. (1)

Auxiliary circle $(x-8)^2 + \left(y - \frac{13}{2}\right)^2 = a^2$

$(2,0)$ lie on it $36 + \frac{169}{4} = a^2$

$$a^2 = \frac{144+169}{4} = \frac{313}{4}$$

$$a = \frac{\sqrt{313}}{2}$$

$$\therefore \frac{\sqrt{313}}{2}.e = \frac{13}{2}$$

$$e = \frac{13}{\sqrt{313}}$$

59. $2\sec^2\alpha - \sec^4\alpha - 2\operatorname{cosec}^2\alpha + \operatorname{cosec}^4\alpha = \frac{15}{4}$

For More Join: @IITJEE_Advanc

$\Rightarrow 2\left(\sec^2\alpha-\operatorname{cosec}^2\alpha\right)+\left(\operatorname{cosec}^2\alpha+\sec^2\alpha\right)\left(\operatorname{cosec}^2\alpha-\sec^2\alpha\right)=\frac{15}{4}$

take $\left(\operatorname{cosec}^2\alpha-\sec^2\alpha\right)$ common in LHS part

$\Rightarrow\left(\operatorname{cosec}^2\alpha-\sec^2\alpha\right)\left(\operatorname{cosec}^2\alpha+\sec^2\alpha-2\right)=\frac{15}{4}$

$\left(\because \operatorname{cosec}^2\theta=1+\cos^2\theta\ \&\ \sec^2\theta=1+\tan^2\theta\right)$

$=4\left(\cot^2\alpha-\tan^2\alpha\right)\left(\cot^2\alpha+\tan^2\alpha\right)=15$

$\left(\because (a-b)(a+b)=a^2-b^2\right)$

$=4\left(\cot^4\alpha-\tan^4\alpha\right)=15$

$\left(\because \cot\alpha=\frac{1}{\tan\theta}\right)$

Apply & take LCM.)

$\Rightarrow 4\left(1-\tan^8\alpha\right)=15\tan^4\alpha$

$=4\tan^8\alpha+15\tan^4\alpha-4=0$

$=4\tan^8\alpha+16\tan^4\alpha-\tan^4\alpha-4=0$

$=\left(4\tan^4\alpha-1\right)\left(\tan^4\alpha+4\right)=0$

$=\tan^4\alpha=\frac{1}{4}$ or $\tan^4\alpha=-4$ (not possible)

$=\tan^2\alpha=\pm\frac{1}{2}$

$\Rightarrow\tan^4\alpha=\pm\frac{1}{2}$

$\Rightarrow\tan^2\alpha=+\frac{1}{2}$

$\left(\because \tan^2\alpha\neq-\frac{1}{2}\right)$

$\Rightarrow\tan\alpha=\pm\frac{1}{\sqrt{2}}$

60. $\begin{vmatrix}2 & 3 & -1\\ 1 & 1 & 1\\ 1 & -1 & |\lambda|\end{vmatrix}=0$

$2\left(|\lambda|+1\right)-3\left(|\lambda|-1\right)-1(-2)=0$

$-|\lambda|+7=0\ \Rightarrow|\lambda|=7$

$\lambda=\pm7$

61. $(3+i)^n=\left(a_n+ib_n\right)$

For More Join: @IITJEE_Advanc

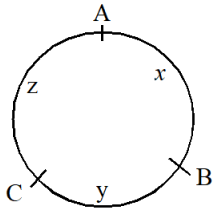
Squaring on both sides

$$(8+6i)^n = a_n^2 - b_n^2 + i(2a_nb_n)$$

$$\sum \frac{a_n b_n}{12^n} = \frac{1}{2}\,\text{Im}\sum_{n=0}^{\infty}\left(\frac{8+6i}{12}\right)^n = \frac{9}{13}$$

62. $$P(E) = \frac{21}{3\times 165} = \frac{7}{165}$$

$$x+y+z = 8$$



$n(S)$ = number of non-negative integral solution

$$= 8 + {}^2C_2 = {}^{10}C_2 = \frac{10\times 9}{2} = 45$$

$n(E)$ = positive integral solution

$${}^7C_2 = 21,\quad P(E) = \frac{21}{45} = \frac{7}{15}$$

63. $$L = \lim_{x\to 4}\left(\frac{\log_5\left(\frac{5x}{4}\right)}{\log_5\left(\frac{20}{x}\right)}\right)^{\frac{1}{\log_5\left(\frac{x}{4}\right)}} ;1^{\infty}\text{ form}$$

$$L = e^{\lim_{x\to 4}\left(\frac{1}{\log_5\left(\frac{x}{4}\right)}\right)\left[\frac{\log_5\left(\frac{5x}{4}\right)}{\log_5\left(\frac{20}{x}\right)}-1\right]}$$

$$L = e^{\lim_{x\to 4}\left(\frac{1}{\log_5 x-\log_5 4}\right)\left[\frac{\log_5\left(\frac{5x}{4}\right)-\log_5\left(\frac{20}{x}\right)}{\log_5\left(\frac{20}{x}\right)}\right]}$$

$$L = e^{\lim_{x\to 4}\left(\frac{1}{\log_5 x-\log_5 4}\right)\left[\frac{\log_5\left(\frac{5x}{4}\times\frac{x}{20}\right)}{\log_5\left(\frac{20}{x}\right)}\right]}$$

$$L = e^{\lim_{x\to 4}\left(\frac{1}{\log_5 x-\log_5 4}\right)\left[\frac{2(\log_5 x-\log_5 4)}{1+\log_5 4-\log_5 x}\right]}$$

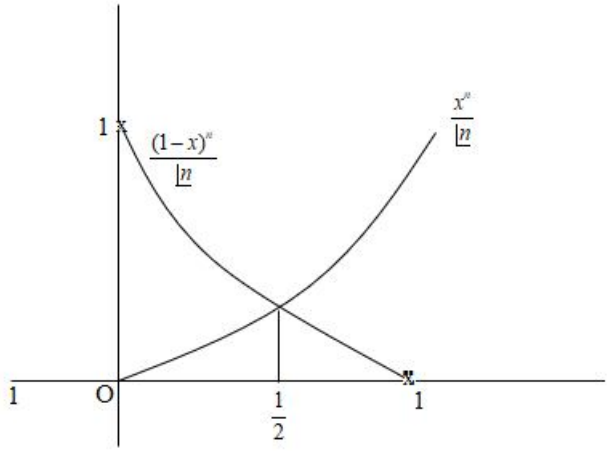
For More Join: @IITJEE_Advanc

$$L = e^{\lim\limits_{x\to 4}\frac{2}{\log_5\left(\frac{20}{x}\right)}}$$

$$L = e^2$$

$$[L] = 7$$

64. $0 \le x \le 1$



$$I_n = \int_0^{\frac{1}{2}} \frac{x^n}{\lfloor n} dx + \int_{\frac{1}{2}}^{1} \frac{(1-x)^n}{\lfloor n} dx$$

$$= \left(\frac{x^{n+1}}{\lfloor n+1}\right)_0^{\frac{1}{2}} - \left(\frac{(1-x)^{n+1}}{\lfloor n+1}\right)_{\frac{1}{2}}^{1}$$

$$= \frac{1}{2^{n+1}.\lfloor n+1} + \frac{1}{2^{n+1}.\lfloor n+1}$$

$$= \frac{1}{2^n.\lfloor n+1}$$

$$\sum_{n=1}^{\infty} I_n = 1 + \frac{1}{2} + \frac{1}{\lfloor 2}\left(\frac{1}{2}\right)^2 + \frac{1}{2^3}.\frac{1}{\lfloor 3} + ...$$

$$= \left(e^{\frac{1}{2}} - \frac{3}{2}\right) \times 2 = 2\sqrt{e} - 3$$

65. $\left(1 + x + x^3 + x^4\right)^n$

$$= \left((1+x)\,(1+x^3)\right)^n$$

$$= (1+x^3)^n\,(1+x)^n$$

$$= \left(1 + {}^nC_1\,x^3 + {}^nC_2\,x^6 + {}^nC_3\,x^9 + {}^nC_4\,x^{12} + {}^nC_5\,x^{15}\right)$$

For More Join: @IITJEE_Advanc

$$\left(1+{}^{n}C_1\,x+{}^{n}C_2\,x^2+{}^{n}C_3\,x^3+.....+{}^{n}C_{15}\,x^{15}\right)$$

$$=\left({}^{n}C_0\,{}^{n}C_{15}+{}^{n}C_1\,.\,{}^{n}C_{12}+{}^{n}C_2\,.\,{}^{n}C_9+{}^{n}C_3\,.\,{}^{n}C_6+{}^{n}C_4\,.\,{}^{n}C_3+{}^{n}C_5\,.\,{}^{n}C_0\right)$$

$$=\sum_{r=0}^{5}{}^{n}C_r\,.\,{}^{n}C_{15-3r}$$

66. $\overrightarrow{OP}\,.\,\overrightarrow{OQ}-\overrightarrow{OR}\,.\,\overrightarrow{OP}=\overrightarrow{OS}\,.\,(\overrightarrow{OQ}-\overrightarrow{OR})$

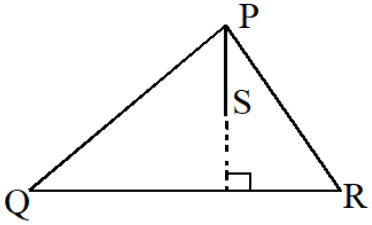
$\Rightarrow \overrightarrow{OP}\,.\,\overrightarrow{QR}=\overrightarrow{OS}\,.\,\overrightarrow{QR}$

$\left(\overrightarrow{OP}-\overrightarrow{OS}\right).\overrightarrow{QR}=0$

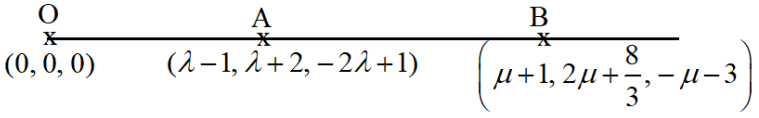
$\overrightarrow{SP}\,.\,\overrightarrow{QR}=0$

$\Rightarrow \overrightarrow{SP}\perp\overrightarrow{QR}$

$\therefore$ S is orthocentre



67.



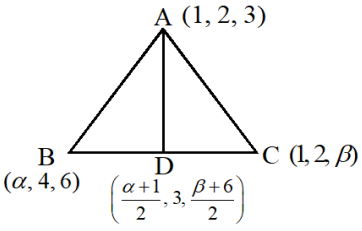
$$\frac{\lambda-1}{\mu+1}=\frac{\lambda+2}{2\mu+8/3}=\frac{-2\lambda+1}{-\mu-3}$$

$$\Rightarrow \lambda=3,\ \mu=\frac{1}{3}$$

$$\therefore A=(2,5,-5),\ B=\left(\frac{4}{3},\frac{10}{3},\frac{-10}{3}\right)$$

$$AB=\sqrt{6}\ \Rightarrow AB^2=6$$

68.



$$\therefore \frac{\frac{\alpha+1}{2}-1}{1}=\frac{1}{1}=\frac{\frac{\beta+6}{2}-3}{1} \qquad \Rightarrow \alpha=3,\ \beta=2 \quad \Rightarrow \alpha+\beta=5$$

For More Join: @IITJEE_Advanc

69. $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$

$C_1 \to C_1 + C_2$

$\begin{vmatrix} 2 & \cos^2\theta & 4\sin 4\theta \\ 2 & 1+\cos^2\theta & 4\sin 4\theta \\ 1 & \sin^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$

$R_2 \to R_2 - R_1,\ R_3 \to 3R_3 - R_1$

$\begin{vmatrix} 2 & \cos^2\theta & 4\sin 4\theta \\ 0 & 1 & 0 \\ 0 & \cos^2\theta & 2+4\sin 4\theta \end{vmatrix} = 0$

$\Rightarrow 2+4\sin 4\theta = 0 \ \Rightarrow \sin 4\theta = \frac{-1}{2}$

$4\theta = n\pi + (-1)^n\left(\frac{-\pi}{6}\right)$

70. $f(x) = x + 3x^2 + 5x^3 + \ldots\ldots + 17x^9$

$x\,f(x) = x^2 + 3x^3 + \ldots\ldots + 15x^9 + 17x^{10}$

$(1-x)\,f(x) = x + 2\left(x^2 + x^3 + \ldots\ldots + x^9\right) - 17x^{10}$

$(1-x)^2\,f(x) = (1-x) + 2x^2(1-x^8) - 17x^{10}\,(1-x)$

$\therefore\ f^1(2) + 2f(2) = 94213$

$= 43 \times 2191$

$2n+1 = 43$

$2n = 42 \qquad n = 21$

71. $\frac{\sum x_i}{6} = 2$ and $\frac{\sum x_i^2}{N} - \mu^2 = 23$

$\alpha + \beta = 10$

$\alpha^2 + \beta^2 = 52$

Solving we get $\alpha = 4, \beta = 6$

$\frac{\sum\left|x_i - \bar{x}\right|}{6} = \frac{5+2+5+8+2+4}{6} = \frac{13}{3} = \frac{m}{n} \Rightarrow m+n = 16$

For More Join: @IITJEE_Advanc